Brain Tumor Detection Using VGG16

Sayantika Chowdhury
School and Computer Science Engineering
Bennett University
Greater Noida, India

***Abstract*—Brain tumors are a complex health problem that presents a significant risk to human life. The timely and accurate detection of brain tumor can affect the effective course of treatment and survival. Typical diagnostic practices (i.e. visual examination of MRI scans) can be time-consuming and prone to human error. To speed up the process using automatic systems, machine learning offers strong analysis methods that can save time and assist radiologists or physicians with an accurate diagnosis. In this research, we describe a machine learning–based system of brain tumor detection using MRI images. The proposed system extracts features utilized from brain scans and applies machine learning algorithms to contribute towards classifying and distinguishing benign, malignant, and normal tissue. In all experiments, the machine learning model yielded high scores of accuracy and performance in correctly identifying tumor type to reduce the opportunity for misdiagnosis. In summary, because the machine learning model is capable of documenting brain tumor identification, we present evidence of the potential use of medical imaging and suggest that machine learning models can be considered a decision-support tool for radiologists and physicians.**

***Index Terms*—Brain Tumor Detection, MRI, VGG16, Deep Learning, Transfer Learning, Medical Imaging, Machine Learning.**

I. Introduction

The human body has trillions of cells that work collaboratively to maintain health and equilibrium. Cells undergo a natural process of growth, division, and death; however, some cells have lost this regulation and can become uncontained and when sufficient abnormal cells have formed a large enough mass, we refer to this as a tumor. Tumors can be benign or malignant; malignant tumors can be fatal for human beings. According to the Central Brain Tumor Registry of the United States (CBTRUS), the prevalence of benign or malignant brain tumors a year is in the tens of thousands, underscoring the magnitude of this health issue taking place all around the world.

Within the medical profession, documentation is essential, and imaging is a significant component of the documentation process. X-ray imaging, Computed Tomography (CT) scans, and Magnetic Resonance Imaging (MRI) techniques are all considered imaging techniques commonly used in this process. MRI is recognized as the best method for detecting abnormalities in the brain because it is able to use less radiation than other imaging and provides a better image of soft tissue. However, and similar to other forms of imaging, the manual reviewing of MRI images by radiologists can be time-consuming, prone to human error, and inefficient when there is large amounts of patient data to review. This reveals an opportunity to create automated systems that will assist radiologists in the interpretation of MRI scans. In the last few years, machine learning and deep learning technologies have been applied to medical image analysis, such as machine learning models trained to classify brain tissue as normal, benign tumor, or malignant tumor based on information derived from MRI scans. Deep learning models, like CNNs and transfer learning, improve results because they can learn from data context and utilize complex patterns to glean features from specifically designed data (i.e., labeled images). Methods of data augmentation, normalization then the organization of labels, further strengthens machine learning methods to resolve the variability in medical images. Features designed for this purpose are provided in the workflow presented in this paper for the detection of a brain tumor from MRI images and classify the tumor as meningioma versus glioma versus pituitary tumor versus no tumor. The workflow leverages preprocessing (i.e., resize, augment, normalize) to prepare the data before the use of a transfer learning model (VGG16), with adaptations for medical use. To further usability of the model for classification, a web application option is provided, using a Flask web application where users can upload MRI images, while classification is provided in real-time via interactivity. This shows how AI can serve as a health decision support tool, particularly providing support to clinicians, through timely diagnosis and reducing the issues around manual interpretation of images. These systems have the ability to develop advanced diagnostic support for clinical practice while increasing accessibility, reliability, and cost-effectiveness, in addition to linking clinical practice to AI research.

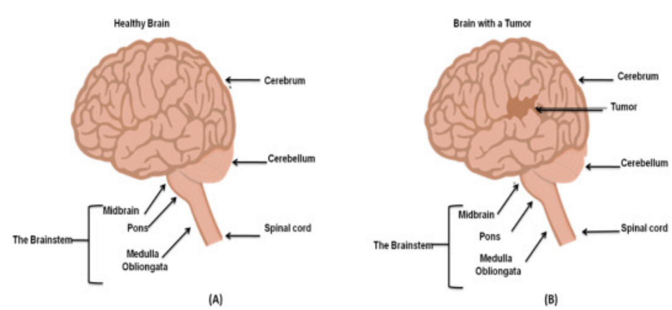


Fig. 1. Comparison between a healthy brain and a brain with a tumor.

II. BACKGROUND

The rapid advancement of deep learning techniques has opened new avenues for medical image analysis, particularly in the detection and classification of brain tumours from MRI scans. A review of existing literature and identification of key research gaps are presented in this section.

A. Literature Review

The authors [1] proposed a deep learning based automated brain tumor detection system by combining a custom CNN model with a VGG-16 transfer-learning architecture. A total of 253 MRI images were used, with techniques like preprocessing, data augmentation, and an 80/10/10 train-validation-test split applied. Both models were first trained and then combined using majority voting for better classification outcomes. In comparison to their individual contributions, their overall performance was much better with the proposed ensemble strategy. These results confirm the fact that the application of deep learning or an ensemble method is reliable and efficient enough for the classification of brain tumors based on MRI scans.

The authors [2] perform a binary classification of brain tumor (tumor vs. no-tumor) in MRI images using the transfer-learning approach with the VGG 16 deep convolutional neural network architecture. Their pipeline includes the acquisition of MRI image data, data preprocessing, such as resizing, normalization, and augmentation, and fine-tuning VGG-16 for the binary classification of healthy versus tumorous brain images. They report competitive accuracy, demonstrating that VGG-16 can serve effectively as a backbone for brain tumor prediction in a binary setting. It is suggested that such classification based on deep learning may help in early detection. The limitations are the reliance on publicly available datasets, which may not reflect full clinical variability, and only binary classification without multi-class tumor type classification. It is recommended that future work should be performed to expand the datasets and number of tumor types and to optimize the models for their clinical deployment.

The author of the paper [3] proposed a hybrid brain-tumor classification method by combining FCM segmentation with the deep-learning model VGG-16. In their method, FCM is applied to segment the tumor regions from MRI images, and then classification is performed using VGG-16. Such a strategy outperforms that when using VGG-16 alone, due to the fact that the tumor region is focused on before classification. The study demonstrates that integrating segmentation and deep learning improves the performance of brain-tumor detection.

The author of the paper [4] proposed the use of deep CNN-based models for the detection of brain tumors on MRI images in a binary/multiclass classification manner. They trained a custom 23-layer CNN and implemented VGG-16-based transfer learning strategies in order to avoid overfitting on small datasets. The framework thus registered a very high accuracy of 97-100% with good generalization compared to previously presented methods. The study itself calls for more diverse clinical datasets as it emphasizes how powerful deep convolutional approaches are in medical image processing.

The author of the paper [6] propose a hybrid deep-learning model that incorporates both CNNs and LSTM memory layers to classify brain tumors from MRI images. They frame it as an IoMT-oriented system: one that preprocess MRI data, feeds it into the CNN-LSTM architecture, and outputs the tumor classification (likely "tumor" vs. "no¬tumor", or various tumor types). The obtained results outperformed traditional CNN models. This study depicts that incorporating temporal/sequential modeling-after feature extraction by CNN-can improve the accuracy of tumor detection and classification. They point to future work involving larger data sets, real-world deployment, and IoMT integration.

The author of the paper [7] propose a ensemble deep learning framework for detection of brain tumors combining several CNN models into one in order to improve accuracy and reliability. The ensemble reduces errors due to limitations in any one model by aggregating the predictions from each architecture. Experiments that they conducted show that the ensemble achieved better classification performance and generalization for MRI datasets. This study highlight how ensembling reinforce robustness in analyse the medical images, although it requires more computational resources and larger datasets for optimum performance.

The author of the paper [8] proposed a CNN base detection system for brain tumors using MRI images, support by fuzzy C-means segmentation to isolate tumor regions. Their CNN model significantly outperformed the traditional classifier like SVM and KNN by achievE an accuracy of about 97%. Their work demonstrate that deep learning performed better in handling variations in shape, size, and intensity of tumors compared to handcrafted methods. They also emphasize developing a large enough and more diverse datasets for their clinical use.

B. Research Gaps

Many studies are looking at classification, which is tumor or no tumor.. In the real world doctors need to know what kind of tumor it is, like meningioma, glioma or pituitary tumors. Some methods use deep learning that needs a lot of computer power so they are not very useful in a hospital setting.T here are not studies that compare different deep learning models in a fair way. So we need to make learning models that are good, at classifying brain tumors like meningioma, glioma and pituitary tumors and can work with different sets of patient data without using too much computer power.

III. PROPOSED METHODOLOGY

The system offer a fully automated deep learning-based model for detecting and classification of brain tumors on MRI scans. The method uses a complete end-to-end pipeline that involves dataset preprocessing, data augmentation, transfer learning using VGG16, and generation of multi-class tumor classification. The technical workflow of the proposed system will be discussed below.

A. Image Preprocessing and Data Augmentation

To improve model acuuracy and reduce overfitting, several preprocessing and augmentation operations are applied:

- **Resizing:** All MRI the images are resized to $128 \times 128 \times 3$ dimensions.
- **Normalization:-** Pixel intensitie scaled to the range of $[0, 1]$.
- **Augmentation:** A custom augmentation module is use that randomly varie image brightness and contrast using controlled ranges.
- **Batch Generator:** A custom generator dynamically load images, applies augmentation, and encode labels for efficient training.

B. Transfer Learning Using VGG16

The VGG16 architecture which was previously train on the ImageNet dataset,is the foundation for the suggested model. The following how transfer learning is use to modify the previously train model for medical imaging:

- The VGG16 model is loaded with `include_top = False`, enabling extraction of convolutional feature maps.
- All convolutional layers are initially frozen to preserve pre-trained weights.
- The last three convolutional layers are selectively unfrozen to allow fine-tuning on MRI data.
- The resulting feature extractor outputs high-level representations suitable for tumor classification.

C. Classification Head Design

On top of the VGG16 feature extractor, a custom classification network is constructed:-

- A Flatten layer convert the extracted feature map into a 1D feature vector.
- A Dropout layer (0.3) reduce overfitting by randomly deactivating neurons.
- A Dense layer with 128 neurons (ReLU activation) performs classification.
- Another Dropout layer (0.2) improves regularization.
- A Softmax output layer with four neurons corresponds to the MRI classes.

The model is compilers use the Adam optimizer with a learning rate of 0.0001 and train 5 epochs using sparse categorical cross-entropy loss.

D. Model Training

The model is train using mini-batch generated by the custom data generator. Each epoch process augmented sample, enable the model to learn variations in MRI imaging conditions. The model achieves:

- Improve accuracy during successive epochs,
- Stable convergence in loss and accuracy,
- Final training accuracy of 97.22%.

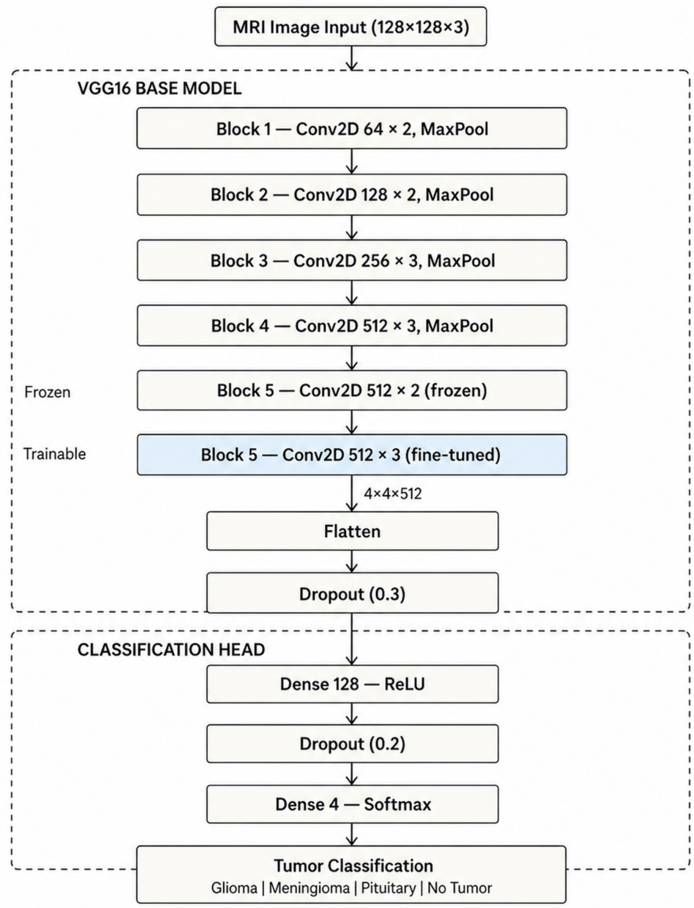


Fig. 2. Block Diagram of the Proposed VGG16-Based Transfer Learning Architecture

E. Model Evaluation

The trained model is evaluates multiple metrics:

- **Classification Report:-** Precision, recall, and F1-score achieve a overall accuracy of 95%.
- **Confusion Matrix:-** Highlight classification performance across tumor classes.
- **ROC–AUC Curves:-** Multi-class ROC confirms strong discriminative capability.

F. Tumor Detection

A prediction module is developed to evaluate new MRI images:

- Loads and preprocesses the image,
- Generates the predicted class and confidence score,
- Displays the MRI with the corresponding prediction,
- Outputs “No Tumor Detected” for non-tumor images.

This pipeline demonstrates the model’s ability to perform accurate real-time tumor detection.

IV. EXPERIMENTS

This section present the performance evaluation of the proposed VGG16-based transfer learning model for brain tumor classification. The results include training behavior, classification accuracy, confusion matrix analysis, ROC–AUC evaluation, and sample prediction outputs.

A. Dataset Description

The Brain Tumor MRI Dataset [16] used in this study was sourced from Kaggle and contains a total of 7,200 human brain MRI images, categorised into four classes: glioma tumour, meningioma tumour, pituitary tumour, and no tumour. These images capture variations in tumour shape, size, and location across different patients, making the dataset well-suited for training a deep learning model for automated tumour detection.

The dataset comes pre-divided into a training set and a testing set, with balanced class distributions maintained across both splits. The training set consists of 5,600 images with 1,400 images per class, while the testing set contains 1,600 images with 400 images per class. Each class is organised into its own subdirectory under the `Training/` and `Testing/` folders, named `glioma/`, `meningioma/`, `pituitary/`, and `notumor/` respectively, making label extraction straightforward during data loading.

The equal distribution across all four classes ensured that the model received fair exposure to each tumour category during training, avoiding any class-frequency bias in the final evaluation results.

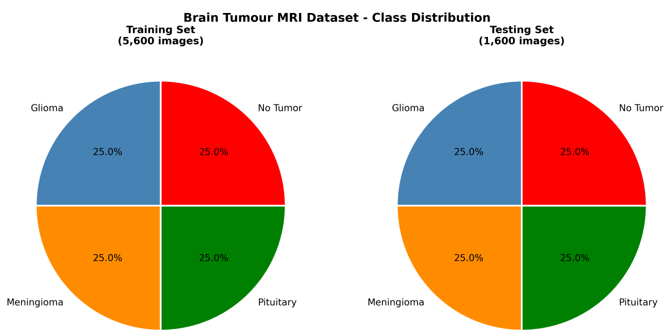


Fig. 3. Class Distribution of the Brain Tumour MRI Dataset across Training and Testing Sets

B. Dataset Preparation

Preparation of the Dataset The MRI dataset is composed of four classes of brain images: glioma, meningioma, pituitary, and healthy (non-tumor). There are training and test subsets of the dataset, and the images were stored in respective subdirectories according to class. Once the images is or- ganized for training and testing their file path and label is collected. Randomization is conducted and organized to eliminated the bias in class ordering, to the ensure randomness, to enhance the generalization the ability of the model during training.

C. Hyper Parameter Details and Tuning

The following hyperparameters and experimental details were used for the VGG16-based Transfer Learning model implemented in this study:

- **Epochs:** 5
- **Batch Size:** 20
- **Loss Function:** Sparse Categorical Cross-Entropy
- **Optimizer:** Adam
- **Learning Rate:** 0.0001
- **Input Image Size:** $128 \times 128 \times 3$ (RGB)
- **Base Model:** VGG16 (pre-trained on ImageNet), with top classification layers excluded (`include_top=False`)
- **Frozen Layers:** All VGG16 layers except the last three (layers -2, -3, -4 set to trainable)
- **Flatten Layer:** Applied after the VGG16 base to convert 3D feature maps to a 1D vector
- **Dropout Rate (Layer 1):** 0.3 (applied after Flatten layer)
- **Dense Layer:** 128 neurons with ReLU activation function
- **Dropout Rate (Layer 2):** 0.2 (applied after Dense layer)
- **Output Layer Activation:** Softmax (multi-class classification over 4 tumor categories)
- **Evaluation Metric:** Sparse Categorical Accuracy
- **Data Augmentation:** Random brightness and contrast adjustment (factor range: 0.8–1.2), with pixel normalization to $[0, 1]$
- **Steps per Epoch:** Computed as $\lfloor$ total training samples / batch size $\rfloor$
- **Dataset Split:** Pre-partitioned training and testing directories (no manual split ratio applied)

These hyperparameters were selected based on established practices for transfer learning with VGG16 on medical image classification tasks. The use of partial fine-tuning—unfreezing only the last three convolutional layers—allowed the model to adapt ImageNet-learned features to MRI brain tumor patterns while preventing overfitting on the relatively small dataset. Dropout regularization at rates of 0.3 and 0.2 further mitigated overfitting, and a low learning rate of 1×10^{-4} ensured stable convergence during training.

D. Training Performance

The model was train 5 epochs using augmented MRI samples. Fig. 4 illustrate the training accuracy and loss curve. The training accuracy continuosly increase, while the loss decrease continuosly over epoch, indicate stable convergence and effective learning.

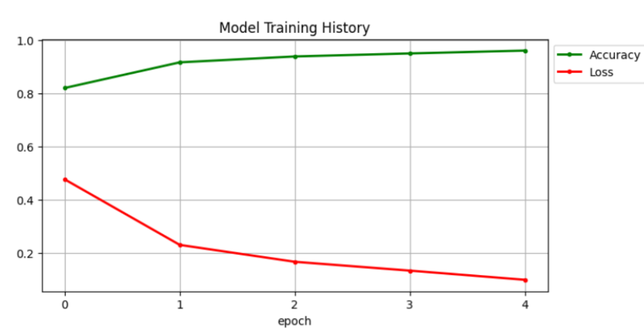


Fig. 4. Model Training History: Accuracy and Loss Across Epochs

E. Prediction Visualization

To validates the inference capability of the model, sample MRI image was tested. The model accurately predict tumor categories with high confidence scores, as shown in Table 1.

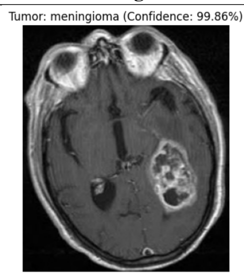
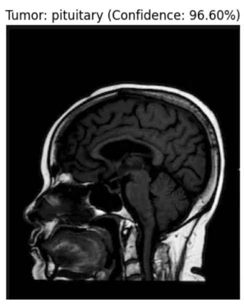
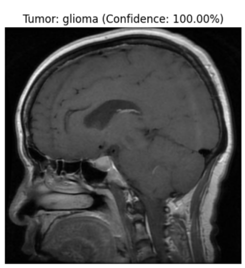
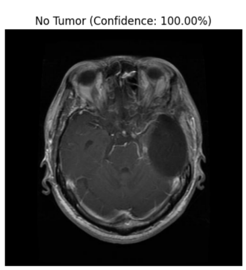
Image	Prediction
	Class: Meningioma Confidence: 99.86%
	Class: Pituitary Confidence: 96.60%
	Class: Glioma Confidence: 100%
	Class: No Tumor Confidence: 100%

TABLE I
SAMPLE PREDICTIONS WITH CONFIDENCE SCORES

F. ROC–AUC Analysis

The ROC curves for all four classes were generated using a one-vs-rest strategy. Fig. 5 shows that each class achieved an AUC of 1.00, demonstrating excellent discrimination ability.

G. Confusion Matrix

Fig. 6 presents the confusion matrix for the model. The diagonal blocks indicate strong predictive performance across all tumor categories. Misclassifications was minimal and occurr mostly among visually similar tumor types.

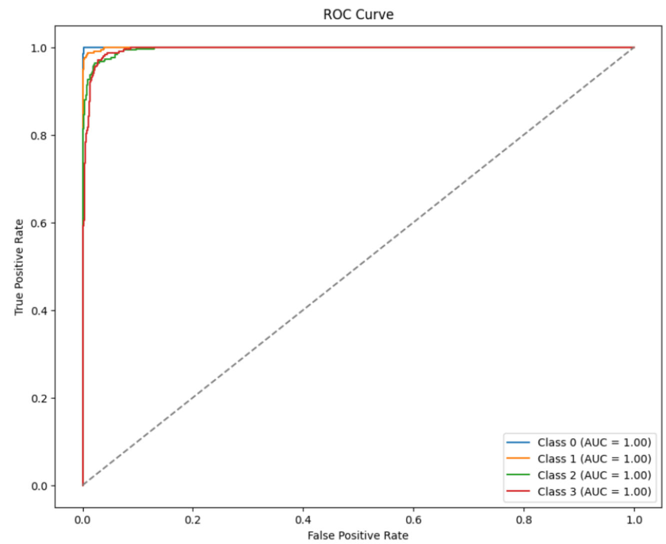


Fig. 5. ROC Curves for Multi-class Brain Tumor Classification

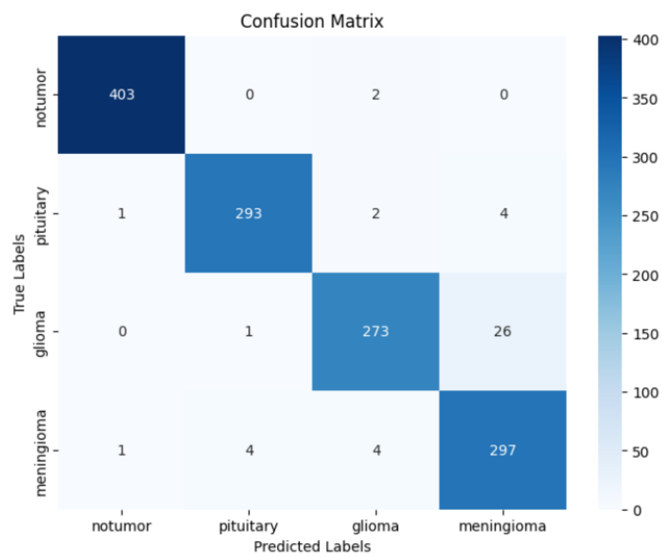


Fig. 6. Confusion Matrix of the Propose Model

H. Results and Discussion

The experimental result demonstrate the robustness of the proposed model. Key findings include:

- Training accuracy reach **97.22%**.
- Test accuracy achieve **95%**.
- High precision, recall, and F1-scores across all four classes.
- AUC values of **1.00** for all tumor categories.
- Accurate tumor detection along with high confident scores is achive in real MRI picture.

TABLE II
CLASS-WISE PERFORMANCE COMPARISON OF PRETRAINED CNN MODELS (%)

Class	Metric (%)	VGG16	ResNet50	MobileNetV2	EfficientNetB0
Glioma	Accuracy	95.75	57.00	76.25	86.25
	Precision	87.15	84.25	94.25	37.75
	Recall	86.25	51.75	76.50	85.00
	F1-score	86.68	64.12	84.34	52.27
Meningioma	Accuracy	96.50	79.25	92.75	25.00
	Precision	82.75	68.75	76.25	37.75
	Recall	86.00	79.12	83.00	2.50
	F1-score	84.34	73.57	79.48	4.69
Pituitary	Accuracy	95.25	89.00	91.25	19.00
	Precision	98.75	86.25	94.00	15.00
	Recall	85.25	90.15	89.00	7.00
	F1-score	91.51	88.11	91.46	9.55
No Tumor	Accuracy	89.75	97.00	95.00	67.00
	Precision	95.25	83.75	91.25	57.00
	Recall	100.00	96.25	97.25	67.00
	F1-score	97.57	89.55	94.17	61.60

Overall, the model has strong classification capacity and reliability for practical deployment for MRI based tumor detection.

I. Model Performance Comparison

Table II presents the performance comparison of multiple pretrained CNN architectures for multi-class brain tumor classification. Among the evaluated models, VGG16 achieved the highest overall classification accuracy of approximately 95%, demonstrating consistent performance across all tumor categories. MobileNetV2 also produced strong results with an overall accuracy of 87.43%, followed by ResNet50 with 79.75% accuracy. In contrast, EfficientNetB0 achieved a comparatively lower accuracy of 38.40%, indicating reduced effectiveness on the given dataset. These results highlight that the VGG16 architecture is the most suitable model for the proposed MRI-based brain tumor classification framework.

V. CONCLUSION

In this work, a deep learning-based framework for brain classification head, and the model showed strong learning. capability and high generalization performance. The experimental results verify the efficiency of the proposed model. In the proposed model, a final training accuracy of 97.22%and an overall testing accuracy of 95% is achieved. with high precision, recall, and F1-scores across all tumor categories. The ROC-AUC values of 1.00 for all classes further confirm the discriminative strength of the model. The Confusion matrix results highlight minimal misclassifications. Indicating excellent performance even among visually similar tumor types. Also, the prediction module gives accurate Tumor identification with high confidence scores on real MRI samples.

Considering everything, the proposed VGG16-based transfer learning system offers a reliable and efficient means of classify brain tumors. With its fast, accurate, and superb detection of tumors, it assists radiologists in making clinical decisions. Future research may focus on developing A complete real-time deployment system for clinical settings, including explainable AI techniques and more. sophisticated deep learning architectures, and growing the dataset.

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